COAXIAL

Fixed Attenuator

VAT-10A+

50Ω Up to 1.7 W DC to 6000 MHz 10 dB Attenuation

THE BIG DEAL

- Wideband Coverage, DC to 6000 MHz
- Up to 1.7 Watt Power Handling
- Rugged Unibody Construction
- Excellent VSWR
- Excellent Flatness



Generic photo used for illustration purposes only

Model No.	VAT-10A+
Case Style	FF704
Connectors	SMA

+RoHS Compliant
The +Suffix identifies RoHS Compliance. See our web site for RoHS Compliance methodologies and qualifications

APPLICATIONS

- Signal Level Adjustment
- Impedance Matching

PRODUCT OVERVIEW

Mini-Circuits' VAT-A series are fixed attenuators from DC to 6000 MHz frequency range with excellent flatness in attenuation. VAT-A series is available with nominal attenuation of 1 to 30 dB. This attenuator series supports testing and measurement applications. Precise performance, excellent VSWR and rugged unibody construction makes the model an ideal solution for systems requiring precise attenuation across a very wide frequency range.

KEY FEATURES

Feature	Advantages
Rugged construction	Excellent durability for a long lifetime of use
Up to 1.7 Watt power handling	Good power handling
Excellent VSWR	Well-matched for 50 Ω systems
Flat attenuation	Good performance over the band

REV. C
ECO-029367
VAT-10A+
EDU4141
MCINU
260630

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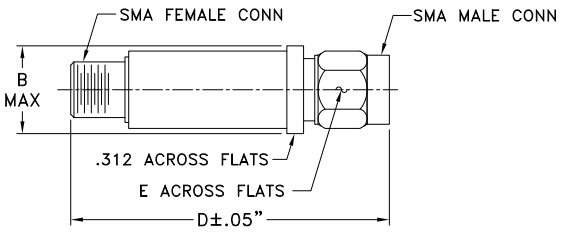
VAT-10A+

MAXIMUM RATINGS

Operating Temperature	-45°C to 100°C
Storage Temperature	-55°C to 100°C

Permanent damage may occur if any of these limits are exceeded.

OUTLINE DRAWING

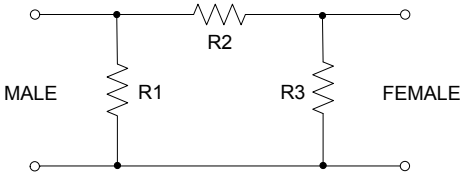


OUTLINE DIMENSIONS (Inch / mm)

B	D	E	wt
.410	1.43	.312	grams
10.41	36.32	7.92	10.0

Note: Please refer to case style drawing for details

ELECTRICAL SCHEMATIC



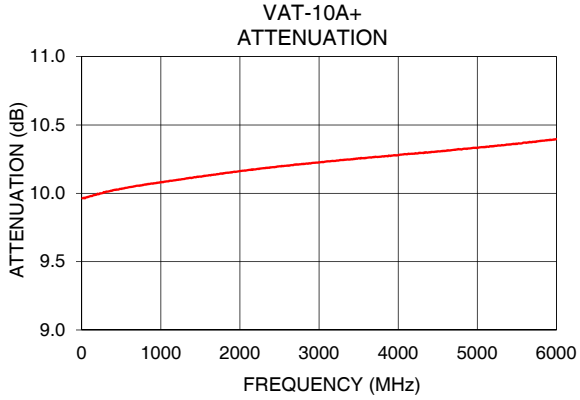
ELECTRICAL SPECIFICATIONS AT 25°C

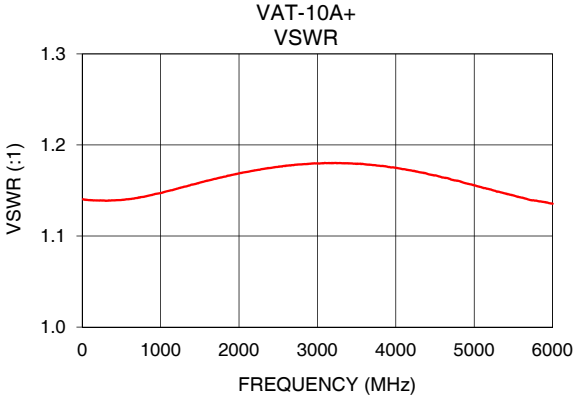
Parameter	Condition (MHz)	Min.	Typ.	Max.	Unit
Frequency Range		DC	-	6000	MHz
Attenuation[1] nominal[3]	10	-	10 ± 0.3	-	dB
Attenuation Flatness[2]	DC-3000	-	0.25	-	dB
	3000-5000	-	0.20	-	
	5000-6000	-	0.15	-	
	DC-6000	-	0.35	-	
VSWR	DC-3000	-	1.2	1.5	:1
	3000-5000	-	1.2	1.5	
	5000-6000	-	1.4	-	
Input Power[4]		-	-	1.7	W

1. Attenuation varies by 0.3 dB max. over temperature.
2. Flatness = variation over band divided by 2.
3. Nominal attenuation at 10 MHz
4. Maximum RF power at 25°C is 1.7 W; Derate linearly to 1.0 W at 85°C

TYPICAL PERFORMANCE DATA

Frequency (MHz)	Attenuation (dB)	VSWR (:1)
10	9.96	1.14
100	9.98	1.14
500	10.03	1.14
900	10.07	1.15
1000	10.08	1.15
1400	10.11	1.16
1500	10.12	1.16
2000	10.16	1.17
2500	10.20	1.18
2800	10.21	1.18
3000	10.23	1.18
4000	10.28	1.17
4500	10.31	1.17
5000	10.33	1.16
6000	10.40	1.14





NOTES

A. Performance and quality attributes and conditions not expressly stated in this specification document are intended to be excluded and do not form a part of this specification document.
B. Electrical specifications and performance data contained in this specification document are based on Mini-Circuit's applicable established test performance criteria and measurement instructions.
C. The parts covered by this specification document are subject to Mini-Circuits standard limited warranty and terms and conditions (collectively, "Standard Terms"); Purchasers of this part are entitled to the rights and benefits contained therein. For a full statement of the standard. Terms and the exclusive rights and remedies thereunder, please visit Mini-Circuits' website at www.minicircuits.com/MCLStore/terms.jsp